
TIMETABLE CONSTRAINTS

HARD CONSTRAINTS (Must be strictly satisfied)

Time & Daily Structure

1. Timetable spans **5 working days**.
 2. Each **lecture** is **1 hour**.
 3. Each **lab session** is **2 hours** (double lecture time).
 4. Each division must have **6–8 academic hours per day**.
 5. Daily hours **must vary** (cannot be same every day).
 6. Each day contains **exactly one 1-hour break**, which:
 - Cannot be the first or last slot
 - Can be **any slot in between**
 7. No lecture or lab can occur during the break.
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Lab Constraints (Corrected)

8. Labs occur **every day** (all 5 days).
 9. Out of 5 days, **exactly 2 days have 2 labs**, the other 3 days have **1 lab** each.
 10. Each lab occupies **2 consecutive hours**.
 11. Each division is split into **2 batches** for labs.
 12. Both batches must run **simultaneously**, in **different lab rooms**.
 13. A lab room can host **only one batch at a time**.
 14. Lab teachers must be available for the **entire 2-hour lab block**.
 15. Labs and theory of the same subject **cannot clash**.
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Teacher Constraints

16. A teacher cannot have **two lectures/labs at the same time**.
 17. Teacher workload per day: **1–3 hours**.
 18. Teachers teaching multiple divisions **cannot have clashes**.
 19. Teachers assigned to a subject must have at least **1 lecture per working day**, if scheduled.
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Subject Constraints

20. Each subject lecture can occur **at most once per day per division**.
 21. Weekly lecture requirements must be satisfied.
 22. Open elective lectures **must not clash** across divisions.
 23. Open electives for a division occupy **a single slot per day** (no splitting).
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Classroom & Resource Constraints

24. There are **3 classrooms per branch** (one per division).
 25. Each classroom can host **only one division at a time**.
 26. Only **3 labs** exist per branch; cannot be double-booked.
 27. All lectures in classrooms, all labs in labs.
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2 SOFT CONSTRAINTS (Optimization Goals)

Student Comfort

1. Avoid more than **2 heavy subjects consecutively**.
2. Prefer labs **before the last two slots** of the day.
3. Balance subject difficulty across the week.
4. Avoid scheduling both earliest and latest slots on the same day.

Teacher Convenience

5. Spread a teacher's workload across the week.
6. Avoid scheduling all lectures on consecutive days.
7. Minimize idle gaps between lectures.

Institutional Preferences

8. Keep same time slots for same subjects across divisions.
 9. Keep open elective slots consistent across divisions.
 10. Balance daily academic load across divisions.
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